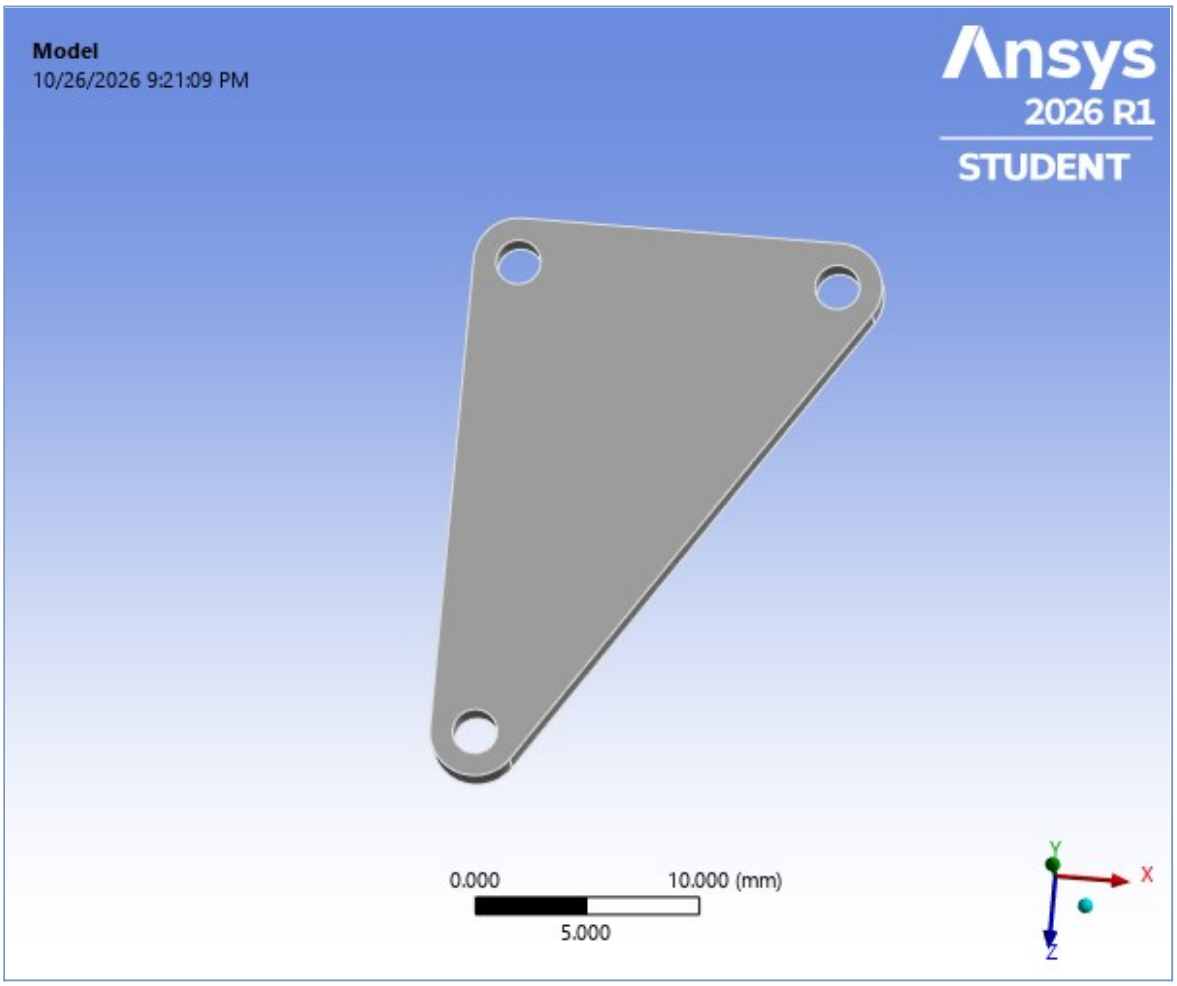
Project*

First Saved	Monday, October 26, 2026
Last Saved	Monday, October 26, 2026
Product Version	2026 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (A4, B4)**
  - Geometry Imports
    - Geometry Import (A3, B3)
  - Geometry
    - SYS\Solid
  - Materials
  - Coordinate Systems
  - Mesh
    - Mesh Controls
  - **Static Structural (A5)**
    - Analysis Settings
    - Loads
    - Solution (A6)
      - Solution Information
      - Equivalent Stress
  - **Structural Optimization (B5)**
    - Analysis Settings
    - Optimization Region
    - Objective
    - Response Constraint
    - Solution (B6)
      - Solution Information
        - Topology Density Tracker
      - Topology Density
- **Material Data**
  - Structural Steel

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, B4)

TABLE 2
Model (A4, B4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4, B4) > Geometry Imports > Geometry Import (A3, B3)

Object Name	*Geometry Import (A3, B3)*
State	Solved

Definition	
Source	D:\Ansys_Temp\WB_User_7460_2\wbnew_files\dp0\SYS\DM\SYS.scdocx
Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4, B4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	D:\Ansys_Temp\WB_User_7460_2\wbnew_files\dp0 \SYS\DM\SYS.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	18.263 mm
Length Y	2. mm
Length Z	25.394 mm
Properties	
Volume	556.91 mm³
Mass	4.3718e-003 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1

Nodes	48196
Elements	9960
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4, B4) > Geometry > Parts

Object Name	*SYS\Solid*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	18.263 mm
Length Y	2. mm
Length Z	25.394 mm

Properties	
Volume	556.91 mm³
Mass	4.3718e-003 kg
Centroid X	6.7129 mm
Centroid Y	1. mm
Centroid Z	-20.322 mm
Moment of Inertia Ip1	0.20717 kg·mm²
Moment of Inertia Ip2	0.26503 kg·mm²
Moment of Inertia Ip3	6.0768e-002 kg·mm²
Statistics	
Nodes	48196
Elements	9960
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:143.175.143	

TABLE 6
Model (A4, B4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4, B4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Mesh

TABLE 8
Model (A4, B4) > Mesh

Object Name	*Mesh*
State	Solved
Display	

Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Mesh Sizing	Adaptive
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	31.343 mm
Average Surface Area	67.496 mm²
Minimum Edge Length	2.0 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (0.1)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	48196
Elements	9960
Show Detailed Statistics	No

TABLE 9
Model (A4, B4) > Mesh > Mesh Controls

Object Name	*Face Sizing*	*Edge Sizing*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	

Geometry	3 Faces	6 Edges
Definition		
Suppressed	No	
Type	Element Size	
Element Size	0.1 mm	
Advanced		
Defeature Size	Default	
Influence Volume	No	
Behavior	Soft	
Bias Type		No Bias
Blend to Neighbors		Yes

Static Structural (A5)

TABLE 10
Model (A4, B4) > Analysis

Object Name	*Static Structural (A5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 11
Model (A4, B4) > Static Structural (A5) > Analysis Settings

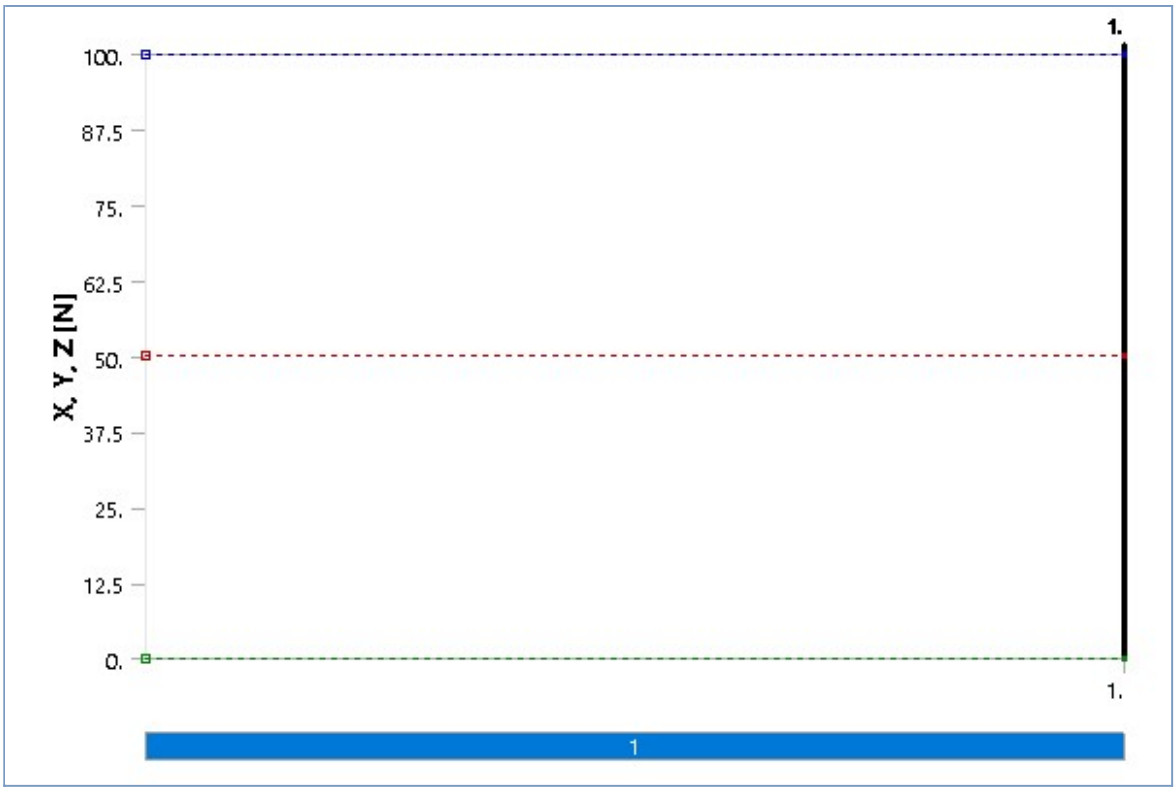
Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Maximum Iterations Per Substep	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled

Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	Yes
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\Ansys_Temp\WB_User_7460_2\wbnew_files\dp0\SYS\MECH\
Future Analysis	Structural Optimization
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	nmm

TABLE 12
Model (A4, B4) > Static Structural (A5) > Loads

Object Name	*Cylindrical Support*	*Bearing Load*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	2 Faces	1 Face
Definition		
Type	Cylindrical Support	Bearing Load
Radial	Fixed	
Axial	Fixed	
Tangential	Free	
Suppressed	No	
Define by		Components
Coordinate System		Global Coordinate System
X Component		50. N
Y Component		0. N
Z Component		100. N

FIGURE 1
Model (A4, B4) > Static Structural (A5) > Bearing Load



Solution (A6)

TABLE 13
Model (A4, B4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	46.81 s
MAPDL Memory Used	497. MB
MAPDL Result File Size	21.562 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 14
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes

Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 15
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Equivalent Stress*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Equivalent (von-Mises) Stress
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	0.21179 MPa
Maximum	85.024 MPa
Average	16.816 MPa
Minimum Occurs On	SYS\Solid
Maximum Occurs On	SYS\Solid
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 2
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

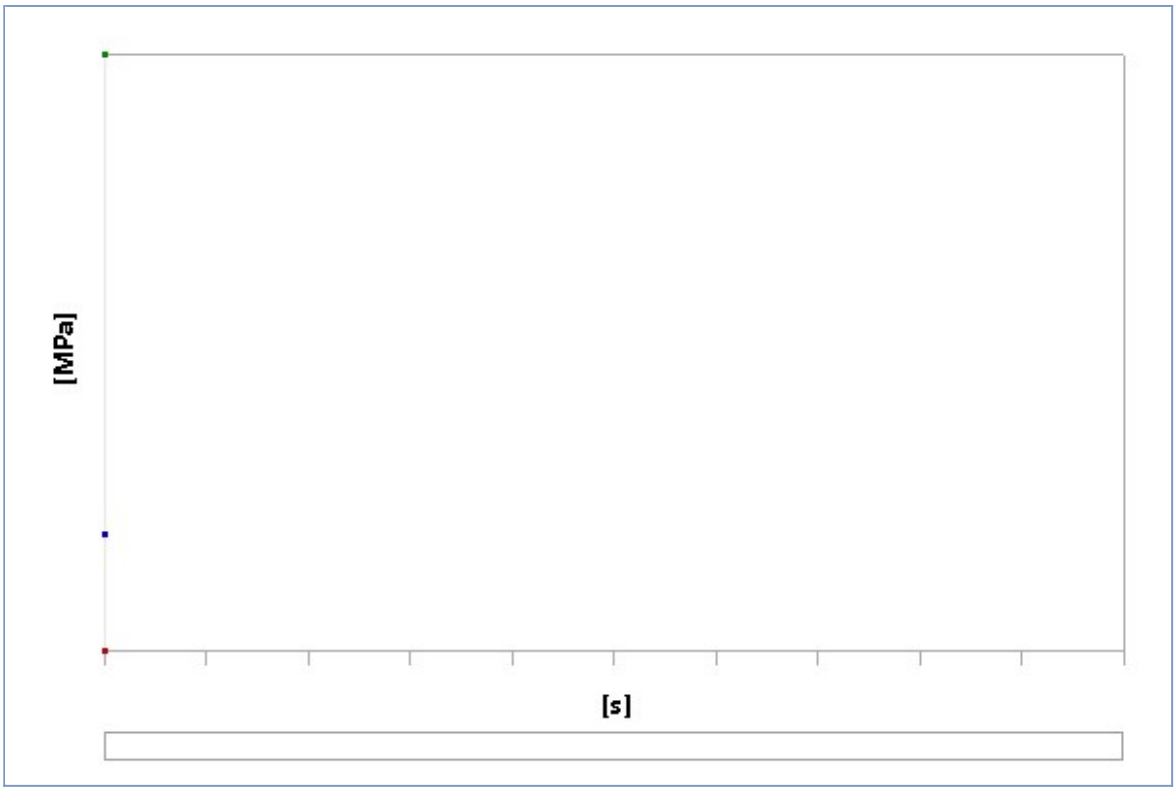
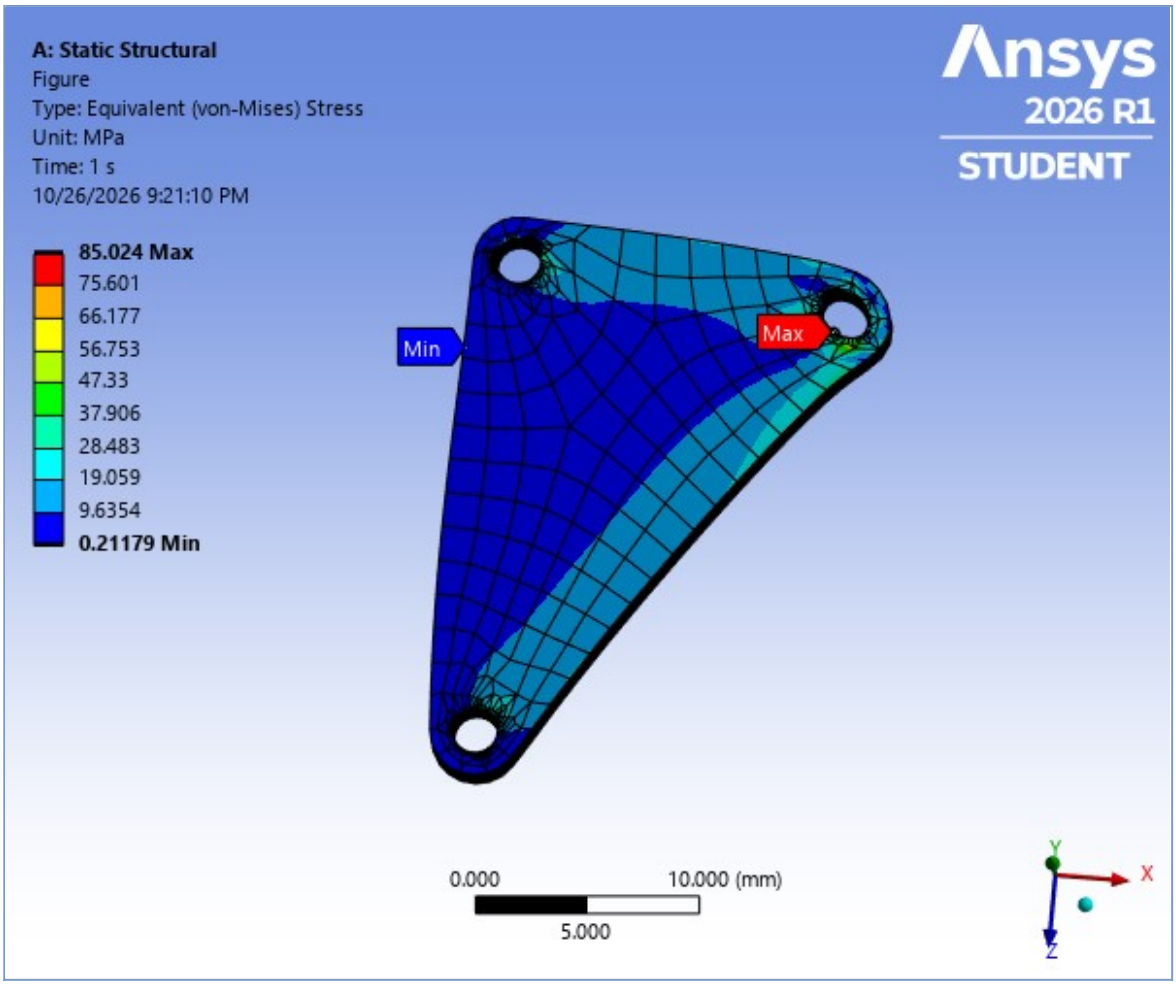


TABLE 16
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	0.21179	85.024	16.816

FIGURE 3
Model (A4, B4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Figure



Structural Optimization (B5)

TABLE 17
Model (A4, B4) > Analysis

Object Name	*Structural Optimization (B5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Structural Optimization
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 18
Model (A4, B4) > Structural Optimization (B5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Definition	
Maximum Number Of Iterations	500.
Convergence Accuracy	0.1 %
Output Controls	
Export Design Properties	No
Store Results At	All Iterations

Solver Controls	
Solver Type	Program Controlled
Algorithm	Program Controlled
Active Set Strategy	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\Ansys_Temp\WB_User_7460_2\wbnew_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	nmm
Max Num Of Intermediate Files	All Iterations

TABLE 19
Model (A4, B4) > Structural Optimization (B5) > Optimization Region

Object Name	*Optimization Region*
State	Fully Defined
Design Region	
Scoping Method	Geometry Selection
Geometry	All Bodies
Exclusion Region	
Define By	Boundary Condition
Boundary Condition	All Boundary Conditions
Exclusion Thickness	Program Controlled
Exclusion Extension	Isotropic
Definition	
Suppressed	No
Optimization Option	
Optimization Type	Topology Optimization - Mixable Density
Initial Volume Fraction	Program Controlled
Penalty Factor (Stiffness)	Program Controlled
Hyperbolic Projection	Program Controlled

TABLE 20
Model (A4, B4) > Structural Optimization (B5) > Objective

Object Name	*Objective*
State	Fully Defined
Definition	
Suppressed	No
Normalized Sum	No

Model (A4, B4) > Structural Optimization (B5) > Objective

Response Type	Goal	Criterion	Formulation	Environment Name	Weight	Multiple Sets	Start Step	End Step	Step	Start Mode	End Mode	Mode
Compliance	Minimize	N/A	Program Controlled	Static Structural	N/A	Enabled	1	1	1	N/A	N/A	N/A

TABLE 21
Model (A4, B4) > Structural Optimization (B5) > Response Constraint

Object Name	*Response Constraint*
State	Fully Defined
Scope	
Scoping Method	Optimization Region
Optimization Region Selection	Optimization Region

Definition	
Type	Response Constraint
Response	Mass
Define by	Constant
Percent to Retain	60 %
Suppressed	No

Solution (B6)

TABLE 22
Model (A4, B4) > Structural Optimization (B5) > Solution

Object Name	*Solution (B6)*
State	Solved
Information	
Status	Done
MAPDL Elapsed Time	6 m 48 s
MAPDL Memory Used	498. MB
MAPDL Result File Size	800.39 KB
Post Processing	
Export Optimal Shape	Only Geometry
-- Topology Result	Topology Density
Definition	
Environment Selection List	A5

TABLE 23
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Objective & Mass Response Convergence
Response Constraint	Response Constraint
Update Interval	2.5 s
Display Points	All

FIGURE 4
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information

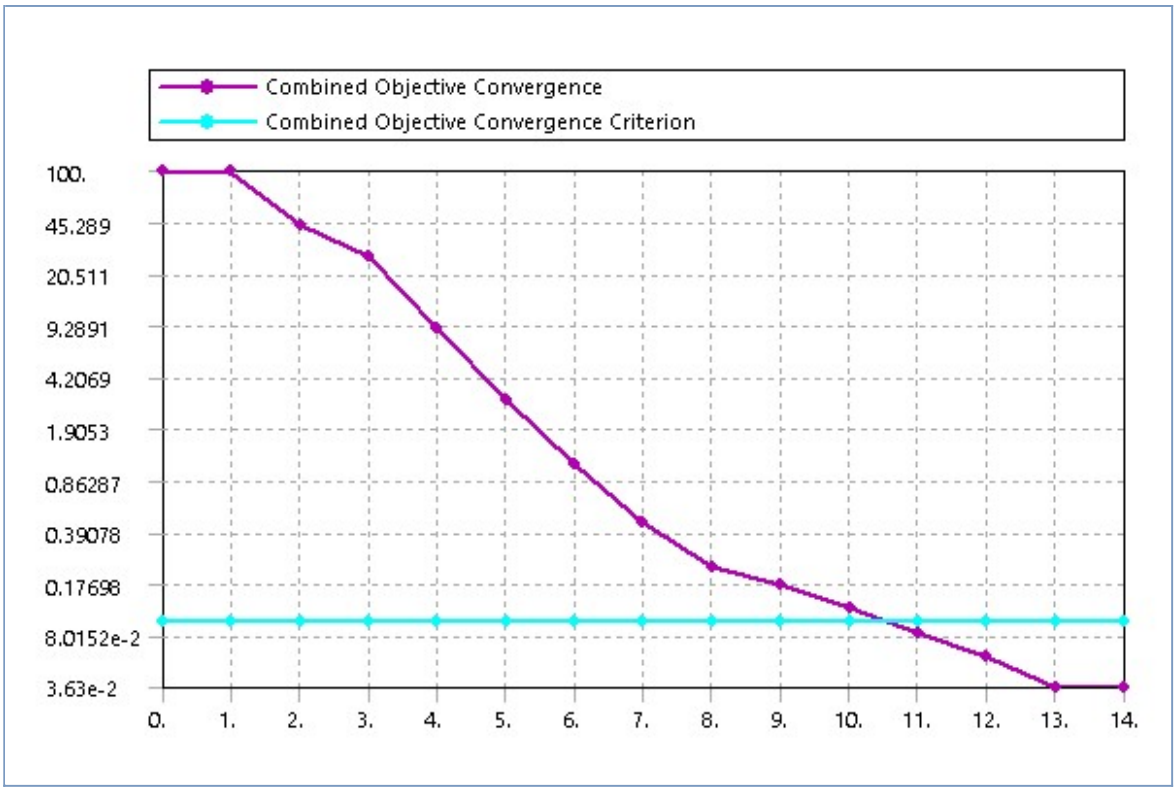


FIGURE 5
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information

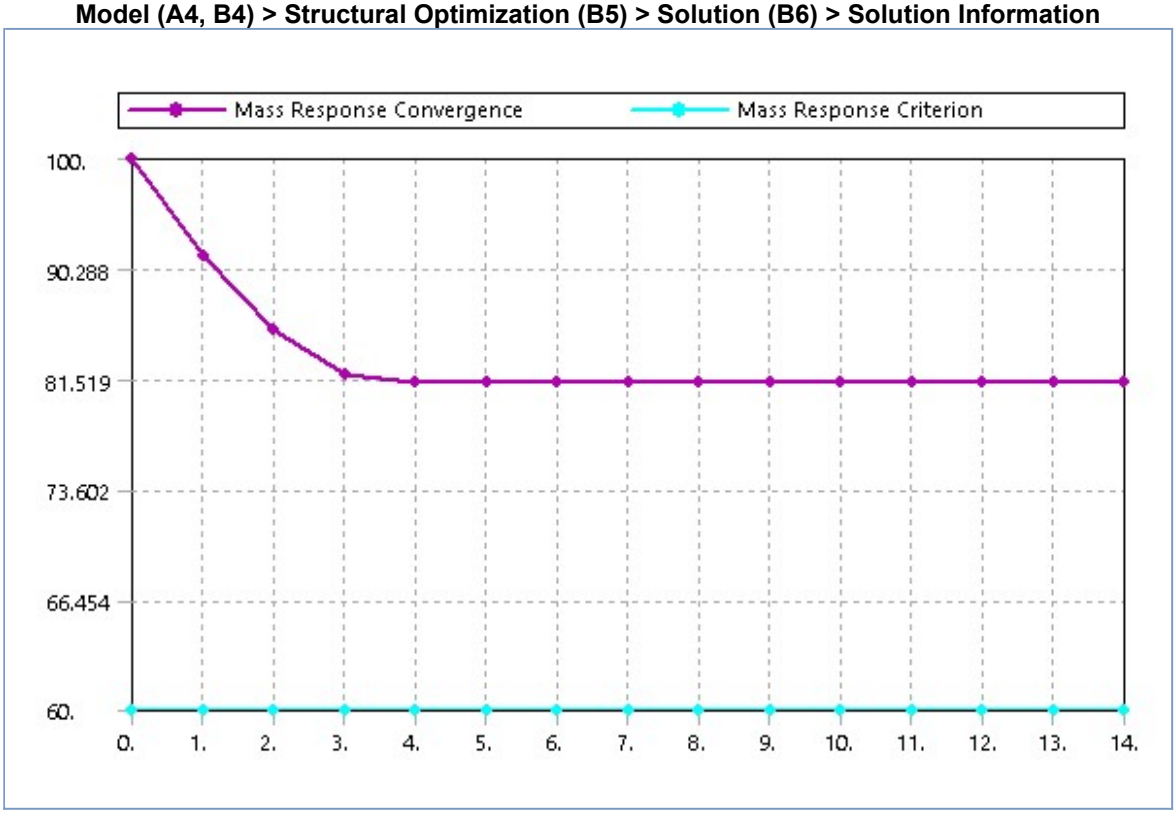


TABLE 24
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Solution Information > Results

Object Name	*Topology Density Tracker*
State	Solved

Scope	
Scoping Method	Optimization Region
Optimization Region	Optimization Region
Definition	
Type	Topology Density Tracker
By	Iteration
Iteration	Last
Retained Threshold	0.5
Separate Data by Entity	No
Suppressed	No
Results	
Minimum	0.
Maximum	1.
Average	0.85194
Visibility	
Show Optimized Region	Retained Region
Information	
Iteration Number	13

TABLE 25
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Results

Object Name	*Topology Density*
State	Solved
Scope	
Scoping Method	Optimization Region
Optimization Region	Optimization Region
Definition	
Type	Topology Density
By	Iteration
Iteration	Last
Retained Threshold	0.5
Exclusions Participation	Yes
Separate Data by Entity	No
Calculate Time History	Yes
Suppressed	No
Results	
Minimum	0.
Maximum	1.
Average	0.85194
Original Volume	556.79 mm³
Final Volume	330.81 mm³
Percent Volume of Original	59.414
Original Mass	4.3708e-003 kg
Final Mass	2.5968e-003 kg
Percent Mass of Original	59.414
Visibility	
Show Optimized Region	Retained Region
Information	
Iteration Number	13

FIGURE 6
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Topology Density

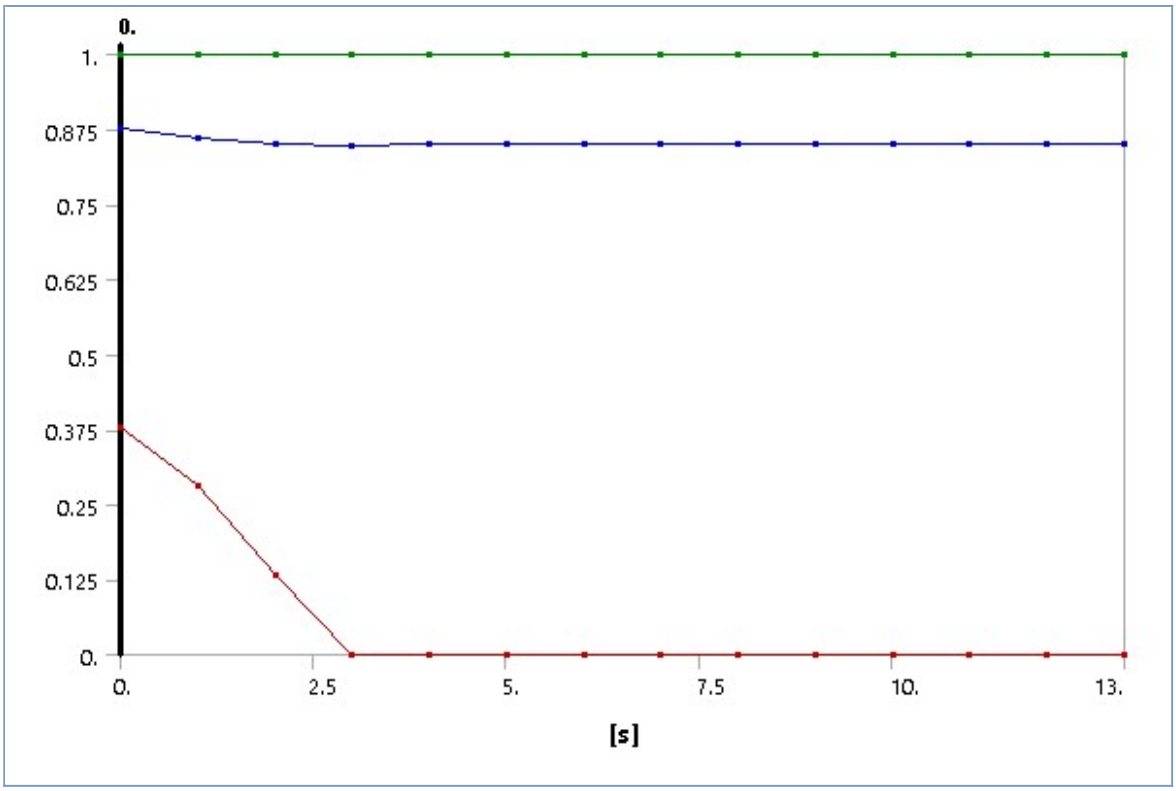
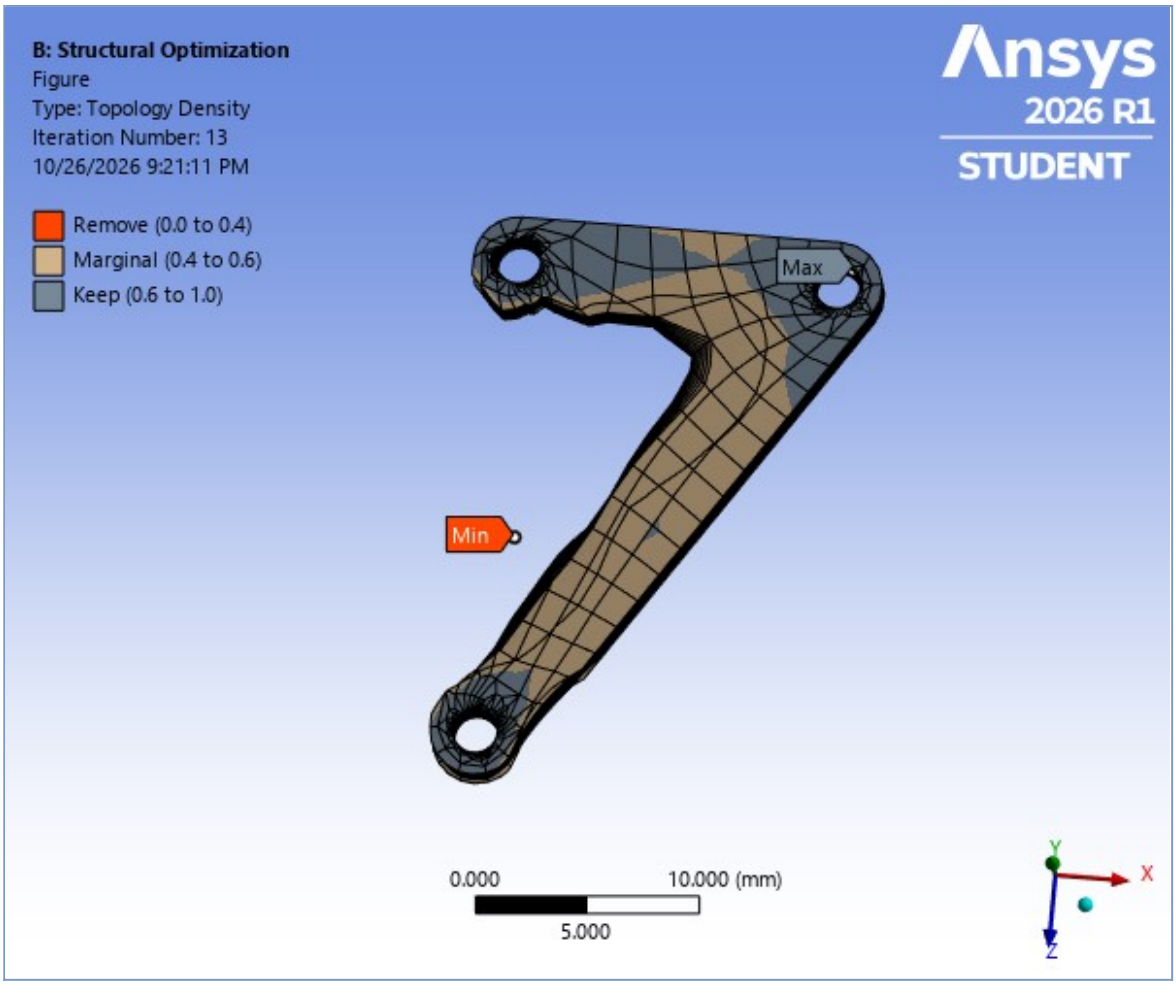


TABLE 26
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Topology Density

Iteration [s]	Minimum	Maximum	Average
0.	0.37967	1.	0.87543
1.	0.27995		0.86075
2.	0.13396		0.85202
3.	1.3833e-003		0.84908
4.	0.		0.85043
5.			0.85102
6.			0.85137
7.			0.85155
8.			0.85163
9.			0.85171
10.			0.85179
11.			0.85185
12.			0.85189
13.			0.85194

FIGURE 7
Model (A4, B4) > Structural Optimization (B5) > Solution (B6) > Topology Density > Figure



Material Data

Structural Steel

TABLE 27
Structural Steel > Constants

Density	7.85e-006 kg mm^-3
Coefficient of Thermal Expansion	1.2e-005 K^-1
Specific Heat	4.34e+005 mJ kg^-1 K^-1
Thermal Conductivity	6.05e-002 W mm^-1 K^-1
Resistivity	1.7e-004 ohm mm

TABLE 28
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 29
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 30
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 31
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 32
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 33
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

TABLE 34
Structural Steel > S-N Curve

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 35
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 36
Structural Steel > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2.e+005	0.3	1.6667e+005	76923	

TABLE 37
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000